

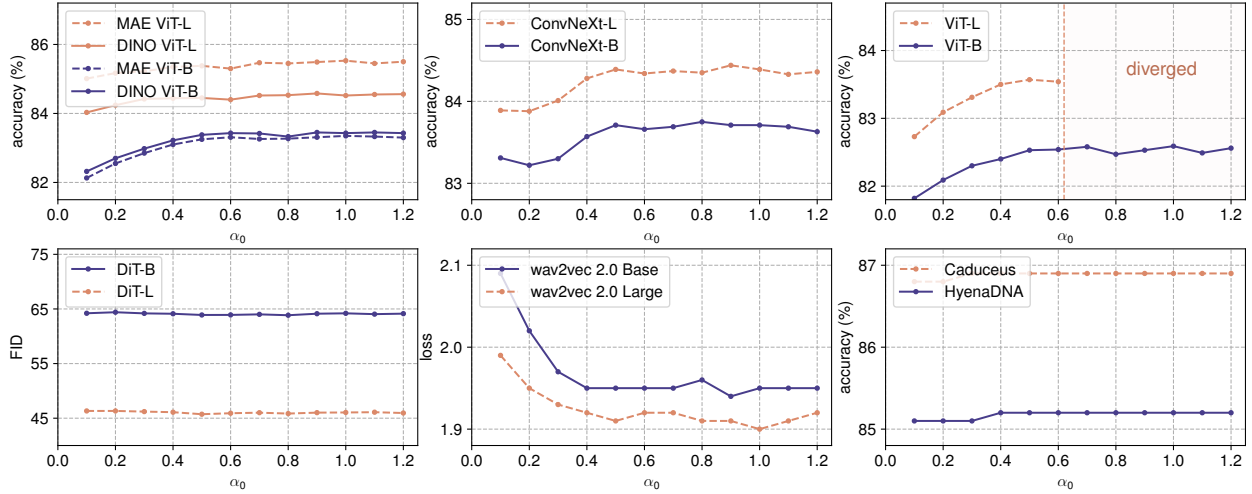


Figure 9 Performance of different tasks across different α_0 values. We benchmark the performance of all non-LLM tasks used in Section 5 with different initial values of α . Performance remains stable across a wide range of α_0 values. The only exception is that supervised ViT-L models (top right panel) will diverge for α_0 values larger than 0.6.

Smaller α_0 results in more stable training. Building on previous observations, we further analyze the factors contributing to training instability. Our findings suggest that increasing either the model size or the learning rate requires lowering α_0 to ensure stable training. Conversely, a higher α_0 requires a lower learning rate to mitigate training instability. Figure 10 shows the ablation of the training stability of supervised ViT with ImageNet-1K dataset. We vary learning rates, model sizes, and α_0 values. Training a larger model is more prone to failure, requiring smaller α_0 values or learning rates for stable training. A similar instability pattern is also observed in LN-based models under comparable conditions, and setting $\alpha_0 = 0.5$ results in a stability pattern similar to that of LN.

Setting $\alpha_0 = 0.5$ as the default. Based on our findings, we set $\alpha_0 = 0.5$ as the default value for all non-LLM models. This setting provides training stability comparable to LN while maintaining strong performance.

7.2 Initialization of α for LLMs

Tuning α_0 enhances LLM performance. As discussed earlier, the default setting of $\alpha_0 = 0.5$ generally performs well across most tasks. However, we find tuning α_0 can substantially improve LLM performance. We tune α_0 across LLaMA models by pretraining each on 30B tokens and comparing their training losses. Table 10 summarizes the tuned α_0 values for each model. Two key findings emerge:

1. **Larger models require smaller α_0 values.** Once the optimal α_0 is determined for smaller models, the search space for larger models can be reduced accordingly.

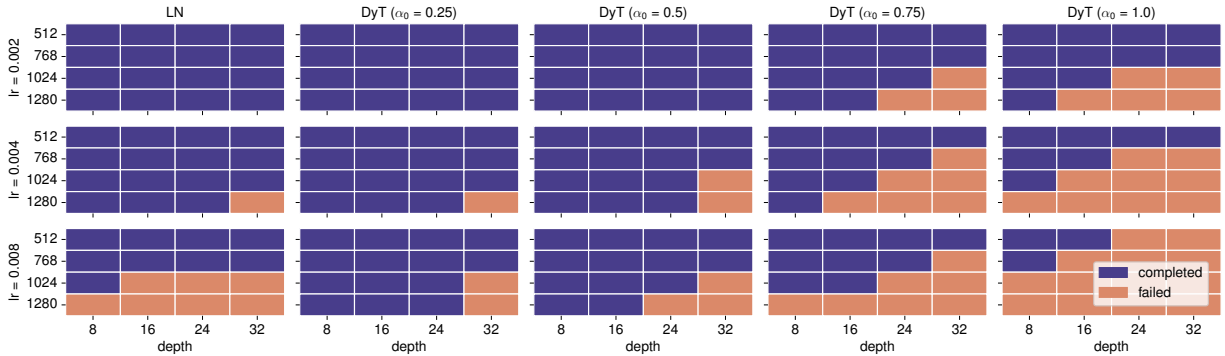


Figure 10 Stability across varying α_0 values, learning rates, and model sizes. We train supervised ViT models on the ImageNet-1K dataset and observe that larger models are more prone to instability for both LN and DyT models. Lowering the learning rate or reducing α_0 enhances stability. LN shows similar stability to DyT with $\alpha_0 = 0.5$.

2. **Higher α_0 values for attention blocks improve performance.** We find that initializing α with higher values for DyT layers in attention blocks and lower values for DyT layers in other locations (i.e., within FFN blocks or before the final linear projection) improves performance.

model	width	depth	optimal α_0 (attention/other)
LLaMA 7B	4096	32	0.8/0.2
LLaMA 13B	5120	40	0.6/0.15
LLaMA 34B	8196	48	0.2/0.05
LLaMA 70B	8196	80	0.2/0.05

Table 10 Optimal α_0 for different LLaMA models. Larger models require smaller α_0 values. We find it is important to initialize α differently in (1) attention blocks (“attention”), versus (2) the FFN blocks, and the final DyT layer before outputs (“other”). α_0 in attention blocks require larger values.

To further illustrate the impact of α_0 tuning, Figure 11 presents heatmaps of loss values of two LLaMA models. Both models benefit from higher α_0 in attention blocks, leading to reduced training loss.

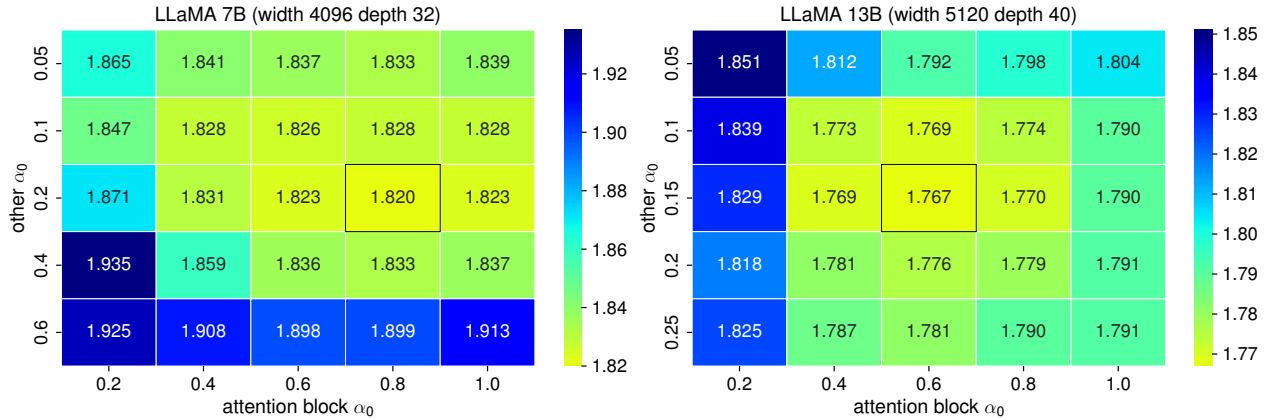


Figure 11 Heatmaps of loss values at 30B tokens for different α_0 settings. Both LLaMA models benefit from increased α_0 in attention blocks.

Model width primarily determines α_0 selection. We also investigate the influence of model width and depth on the optimal α_0 . We find that the model width is critical in determining the optimal α_0 , while model depth has minimal influence. Table 11 shows the optimal α_0 values across different widths and depths, showing that wider networks benefit from smaller α_0 values for optimal performance. On the other hand, model depth has negligible impact on the choice of α_0 .

As can be seen in Table 11, the wider the network, the more uneven initialization for “attention” and “other” is needed. We hypothesize that the sensitivity of LLM’s α initialization is related to their excessively large widths compared to other models.

8 Related Work

Mechanisms of Normalization layers. There has been a rich line of work investigating normalization layers’ role in enhancing model performance through various mechanisms. These include stabilizing gradient flow during training (Balduzzi et al., 2017; Daneshmand et al., 2020; Lubana et al., 2021), reducing sensitivity to weight initialization (Zhang et al., 2019; De and Smith, 2020; Shao et al., 2020), moderating outlier eigenvalues (Bjorck et al., 2018; Karakida et al., 2019), auto-tuning learning rates (Arora et al., 2018; Tanaka and Kunin, 2021), and smoothing the loss landscape for more stable optimization (Santurkar et al., 2018). These earlier works focused on studying batch normalization. Recent studies (Lyu et al., 2022; Dai et al., 2024; Mueller et al., 2024) further highlight the connection between normalization layers and sharpness reduction, which contributes to better generalization.

width / depth	8	16	32	64
1024	1.0/1.0	1.0/1.0	1.0/1.0	1.0/1.0
2048	1.0/0.5	1.0/0.5	1.0/0.5	1.0/0.5
4096	0.8/0.2	0.8/0.2	0.8/0.2	0.8/0.2
8192	0.2/0.05	0.2/0.05	0.2/0.05	0.2/0.05

Table 11 Optimal α_0 (attention / other) across model widths and depths in LLaMA training. Model width significantly impacts the choice of α_0 , with wider networks requiring smaller values. In contrast, model depth has negligible influence.

Normalization in Transformers. With the rise of Transformer (Vaswani et al., 2017), research has increasingly focused on layer normalization (Ba et al., 2016), which has proven particularly effective for sequential data in natural language tasks (Nguyen and Salazar, 2019; Xu et al., 2019; Xiong et al., 2020). Recent work (Ni et al., 2024) reveals that layer normalization introduces strong non-linearity, enhancing the model’s representational capacity. Additionally, studies (Loshchilov et al., 2024; Li et al., 2024) demonstrate that modifying the location of normalization layers within Transformers can improve convergence properties.

Removing normalization. Many studies have explored how to train deep models without normalization layers. Klambauer et al. (2017) introduce an alternative activation function that enables self-normalizing behavior, eliminating the need for explicit normalization. Other works (Zhang et al., 2019; De and Smith, 2020; Bachlechner et al., 2021) propose specialized initialization schemes to stabilize training in the absence of normalization. The pioneering work by Brock et al. (2021a,b) show that high-performing ResNets can be trained without normalization (Smith et al., 2023) through combination of initialization techniques (De and Smith, 2020), weight normalization (Salimans and Kingma, 2016; Huang et al., 2017; Qiao et al., 2019), and adaptive gradient clipping (Brock et al., 2021b). Additionally, their training strategy incorporates extensive data augmentation (Cubuk et al., 2020) and regularization (Srivastava et al., 2014; Huang et al., 2016). The studies above are based on various ConvNet models.

In Transformer architectures, He and Hofmann (2023) explore modifications to Transformer blocks that reduce reliance on normalization layers and skip connections. Jha and Reagan (2024) introduce AERO, a Softmax-only LLM that improves inference efficiency and privacy with minimal performance loss. Alternatively, Heimersheim (2024) propose a method to gradually remove LN from pretrained networks by fine-tuning the model after removing each normalization layer. Unlike previous approaches, DyT requires minimal modifications to both the architecture and the training recipe. Despite its simplicity, DyT achieves stable training and comparable performance.

9 Limitations

Our experiments focus on networks using LN or RMSNorm because of their popularity in Transformers and other modern architectures. Preliminary experiments (see Appendix D) indicate that DyT struggles to replace BN directly in classic networks like ResNets. It remains to be studied in more depth whether and how DyT can adapt to models with other types of normalization layers.

Furthermore, although DyT is conceptually and computationally simpler, we find that DyT offers no speedup over models with normalization layers when properly compiled/optimized (see Appendix C). Its computational benefits across different hardware platforms or deployment environments remain uncertain.

10 Conclusion

In this work, we demonstrate modern neural networks, in particular Transformers, can be trained without normalization layers. This is done through Dynamic Tanh (DyT), a simple replacement for traditional normalization layers. It adjusts the input activation range via a learnable scaling factor α and then squashes the extreme values through an S -shaped tanh function. Although a simpler function, it effectively captures the behavior of normalization layers. Under various settings, models with DyT match or exceed the performance of their normalized counterparts. The findings challenge the conventional understanding of the necessity of normalization layers in training modern neural networks. Our study also contributes to understanding the mechanisms of normalization layers, one of the most fundamental building blocks in deep neural networks.